

Question No 4:

1) Problem Analysis Chart:

Input	Processing	Output	Decision
• Quantity (q)	• Valid input i.e. $q, p > 0$ and $0 \leq d, f \leq 100$	• Error message (if input valid)	• Validation rule
• Price list (p)			$\Rightarrow q > 0$
(q)		• Generated Bill	$\Rightarrow p > 0$
• Discount (d)	• if invalid display error	Discount entry $\rightarrow$ Quantity of Price $\rightarrow$ Subtotal	$\Rightarrow$ discount must be b/w 0 and 100
• Tax percentage (t)			$\Rightarrow$ Tax must be b/w 0 & 100
• Tax percentage	• Calculate subtotal $s = q \times p$	$\Rightarrow$ Discount Amount $\rightarrow$ Discounted subtotal $\rightarrow$ Tax amount	
	• Calculate discount amount $a = s - \frac{s \times d}{100}$	$\Rightarrow$ Final Bill	
	• Calculate final Bill $Billed = a + \frac{a \times t}{100}$		
	• Generate Bill		

2. Input Processing Output:

Input	Processing	Output
• q (Quality)	1) Input and validate Read p, q, d, T	Invalid input Error
• p (price/unit)	if $q \leq 0$ or $p \leq 0$ or $d \leq 0$ or $d > 100$ or $T < 0$ or $T > 100$	Valid input
• d (Discount)	then	
• T (Tax %)	Display specific error and Terminate	• s (subtotal)
		• disc amount
	2) Calculations	
	a) $s = q \times p$	• a (discounted amount)
	b) disc amount = $s \times d$	
	c) Total amount = $s - \text{disc amount}$	• Tax amount
	d) Tax amount = $a \times T$	
	e) $f = a + \text{Tax amount}$	• f (Final Bill amount)

3. Algorithm:

1) Start

2) Input q, p, d, r

3) check ~~until~~

if $(q < 0, p < 0, d < 0 \text{ OR } d \geq 100, r < 0 \text{ OR } r > 100)$
print invalid input

EXIT

END If

4)

ELSE

- Compute Subtotal: $s = q \times p$

- Compute Discant Amount: $\text{disc_amt} = \frac{s \times d}{100}$

- Compute Discanted subtotal, $a = s - \text{disc_amt}$

- Compute Tax Amount: $\text{Tax_amt} = a \times r$

- Compute Final Bill: $f = a + \text{Tax_amt}$

END ELSE

5) Print $p, q, s, \text{disc_amt}, a, \text{Tax_amt}$ & f

6) END

4. Pseudocode:

BEGIN

Declare P, q, T, d as integer

Print "Enter quantity of data item"

Read q

Print "Enter price of item"

Read P

Print "Enter Tan"

Read T

Print "discount"

Read d

If $(P > 0, q > 0, T \geq 0 \text{ AND } T \leq 100, d \geq 0 \text{ AND } d \leq$

Set $s = P * q$, Print s

Set $\text{dis_amt} = (s * d) / 100$, Print dis_amt

Set $a = s - \text{dis_amt}$, Print a

Set $\text{Tax_amt} = a * T / 100$, Print Tax_amt

Set $\text{final Bill, } f = a + \text{Tax_amt}$, Print final Bill

Else

END

5. Flow chart.

